This ICCV paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

PS3: A Multimodal Transformer Integrating Pathology Reports with Histology Images and Biological Pathways for Cancer Survival Prediction

Manahil Raza Ayesha Azam Talha Qaiser Nasir Rajpoot
University of Warwick, UK
{manahil.raza, ayesha.azam, talha.qaiser, n.m.rajpoot}@warwick.ac.uk

Abstract

Current multimodal fusion approaches in computational oncology primarily focus on integrating multi-gigapixel histology whole slide images (WSIs) with genomic or transcriptomic data, demonstrating improved survival prediction. We hypothesize that incorporating pathology reports can further enhance prognostic performance. Pathology reports, as essential components of clinical workflows, offer readily available complementary information by summarizing histopathological findings and integrating expert interpretations and clinical context. However, fusing these modalities poses challenges due to their heterogeneous nature. WSIs are high-dimensional, each containing several billion pixels, whereas pathology reports consist of concise text summaries of varying lengths, leading to potential modality imbalance. To address this, we propose a prototype-based approach to generate balanced representations, which are then integrated using a Transformer-based fusion model for survival prediction that we term PS3 (Predicting Survival from Three Modalities). Specifically, we present: (1) Diagnostic prototypes from pathology reports, leveraging self-attention to extract diagnostically relevant sections and standardize text representation; (2) Histological prototypes to compactly represent key morphological patterns in WSIs; and (3) Biological pathway prototypes to encode transcriptomic expressions, accurately capturing cellular functions. PS3, the three-modal transformer model, processes the resulting prototype-based multimodal tokens and models intra-modal and cross-modal interactions across pathology reports, WSIs and transcriptomic data. The proposed model outperforms state-of-the-art methods when evaluated against clinical, unimodal and multimodal baselines on six datasets from The Cancer Genome Atlas (TCGA). The code will be released upon publication.

1. Introduction

Patient prognosis in oncology refers to the predicted progression and outcome of cancer, often determined by clinical, pathological and molecular factors [44, 71]. It is essential to guide treatment strategies, inform risk assessment and facilitate patient-centered decision-making to optimize survival and quality of life [13, 38]. Recently, integrating multiple data modalities have been shown to improve survival prediction in cancer research, with most studies focusing on combining genomic data such as transcriptomic data with histology whole slide images (WSIs) [7, 52, 54]. These approaches leverage the strengths of each modality—genomics captures molecular subtypes, while WSIs provide spatial and morphological insights into tumor characteristics [72]. However, pathology reports remain an under-utilized resource despite providing essential information to clinicians for estimating the prognosis of cancer patients. Generated by expert pathologists, they are mandatory for definitive cancer diagnosis [42] and contain critical clinical and prognostic information including biomarker status, tumor grading, staging and histological subtypes [24, 27, 41]. They often complement histology and genomic data by incorporating expert-driven diagnostic insights [1, 53]. Moreover, as routinely produced components of clinical workflows, pathological reports serve as a readily available source of medical knowledge [23]. Despite their clinical significance, they remain largely untapped in computational survival prediction models.

Early multimodal approaches often relied on late fusion techniques [6, 31, 64], which integrated unimodal representations only at the final decision stage [12]. However, these methods struggled to model cross-modal interactions, which may be crucial for accurate prognosis. In contrast, more recent studies have investigated early fusion approaches for integrating WSIs and genomic data. Although early fusion captures fine-grained cross-modal interactions more effectively [5, 20, 51, 67], it remains computationally demanding due to the high dimensionality and heterogeneity of the data.

Another key challenge in early fusion of WSIs, genomics and pathology reports is modality imbalance, arising from differences in structure and scale of the data modalities. WSIs are gigapixel-scale images typically divided into thousands of high-dimensional patches containing spatial and morphological information [46]. To process these vast amounts of image data, Multiple Instance Learning (MIL) [11, 63] is commonly employed, allowing models to extract and aggregate relevant features from individual patches. In contrast, pathology reports provide concise textual summaries with significantly fewer raw data points. Further, genomic data, such as RNA-Seq expression levels are typically represented as isolated scalar values, lacking contextual information about their biological functions. This imbalance can lead to modality dominance, where data-rich modalities (*e.g*. WSIs) can disproportionately influence the model compared to modalities with fewer tokens (*e.g*. pathology reports). Addressing this imbalance is crucial for developing an effective multimodal fusion framework that ensures fair contributions from all modalities and improves predictive performance.

To address these challenges, we propose a prototype-based multimodal fusion framework that standardizes and balances representations across three modalities. The prototypes represent large, variable-length inputs as compact embeddings, enabling more effective multimodal fusion. We exploit the inherent morphological redundancy in WSIs by identifying recurring histological patterns within tissue patches [62]. Inspired by previous works [25, 50, 51], we use a Gaussian Mixture Model (GMM) to represent slides, with each mixture component corresponding to a distinct *histological prototype*. This clustering of similar morphological patterns enables compact WSI representations using key histological prototypes. Pathology reports, which contain significantly less raw data than WSIs, often exhibit unstructured or semi-structured formats, inconsistent formatting, varying lengths and levels of detail [41, 47, 59]. Rather than truncating their content, we construct *diagnostic prototypes*—standardized representations designed to capture the most diagnostically relevant sections of each report using self-attention-based mechanisms [60]. To ensure a biologically meaningful representation of transcriptomic data, we draw inspiration from prior methods [20, 51, 72] and transform the values into a set of 50 Cancer Hallmark *pathway prototypes* [33]. This representation aligns gene expression patterns with well-established cellular processes, providing a structured and interpretable genomic feature set.

We propose a multimodal transformer-based framework **PS3**, designed to **P**redict **S**urvival by integrating information from three (**3**) different data modalities: WSIs, transcriptomics and pathology reports. By transforming raw data into compact prototype-based representations, we reduce the disparity in token counts across modalities, achieving a more balanced token distribution. This allows our transformer-based method to effectively model self-attention and cross-attention mechanisms, thereby facilitating the fusion of complementary information from all three modalities. As a result, our approach enhances predictive performance, leading to robust survival prediction. We evaluate the proposed method on six cancer cohorts from The Cancer Genome Atlas (TCGA) [65] and demonstrate that it outperforms both unimodal and multimodal baselines.

To summarize, our main contributions are: (1) a novel framework for representing pathology reports, WSIs and transcriptomic profiles using diagnostic, histological and pathway prototypes, respectively; (2) development of a multimodal transformer, PS3, to predict survival using three modalities; and (3) comprehensive experiments and ablation studies on six cancer cohorts, showcasing the effectiveness of PS3 for cancer survival prediction.

2. Related Work

2.1. Survival Prediction with Whole Slide Images

Given the large number of patches in WSIs, many methods adopt MIL-based methods for efficient processing and analysis [2]. MIL typically involves three key steps: (1) dividing the WSI into hundreds or thousands of smaller patches (or instances), (2) extracting patch embeddings using a feature encoder and (3) aggregating the patch embeddings to obtain a slide-level representation [63]. For survival prediction, various MIL-based approaches have been explored, including graph-based [4, 30, 35] and attention-based MIL models [18, 70] both of which aim to capture global WSI representations for survival analysis. More recently, transformer-based architectures leveraging self-attention mechanisms [10, 21, 60] have been introduced, including hierarchical transformer architectures designed to capture multi-scale WSI representations [49, 68].

2.2. Multimodal Survival Prediction

While unimodal methods have demonstrated strong prognostic capabilities, multimodal approaches have led to further improvements in survival prediction. Most existing multimodal fusion methods integrate genomic or transcriptomic data with multi-gigapixel histology WSIs [6]. Early studies primarily relied on late fusion using techniques such as concatenation [39], Kronecker product [3, 64] or factorized bilinear pooling [31, 45]. However, since late fusion merges unimodal representations only at the final stage [12], it fails to capture cross-modal interactions, leading to sub-optimal integration. Similarly, text-based fusion using concatenation [15] and weighted sum [40] to combine patient health records and medical images suffer from similar limitations.

In contrast, early fusion approaches have gained signif-

icant attention for their effectiveness in modeling cross-modal interactions. Many models utilize transformers [66], incorporating co-attention [5, 34, 74] and cross-attention mechanisms [9, 20, 37, 51] to enhance feature integration. Some extend transformer architectures with hierarchical transformers [29] while others employ prototyping techniques to reduce data dimensionality before fusion, improving efficiency and minimizing the number of tokens processed [50]. Additionally, other approaches such as optimal transport-based methods [51, 67], information bottlenecking techniques [72] and graph-based methods [73] have been explored as alternatives to transformers.

3. Methodology

We introduce PS3, a prototyping-based multimodal framework for survival prediction that integrates histology images, transcriptomic data and pathology reports. We explain the construction of the diagnostic prototypes, histological prototypes and pathway prototypes in Sections 3.1, 3.2 and 3.3, respectively. Section 3.4 describes the Transformer-based multimodal approach, while Section 3.5 outlines the post-fusion processing steps taken for survival prediction.

3.1. Diagnostic Prototypes from Pathology Reports

Feature Extraction: Each pathology report (t), denoted as R_i, is first divided into smaller text segments or sections: $R_i = \{x_1^t, x_2^t \dots x_{N_{t,i}}^t\}$, where $N_{t,i}$ is the number of segments in the ith report. These text segments are tokenized and subsequently passed through a pre-trained and frozen text encoder E_T which extracts feature embeddings for each text segment: $\mathbf{h}_i^t = E_T(x_i^t) \in \mathbb{R}^{d_t}$. The set of embeddings for the ith pathology report are then defined as: $\mathbf{H}_i^t = \{\mathbf{h}_1^t, \mathbf{h}_2^t \dots \mathbf{h}_{N_{t,i}}^t\}$.

Self-Attention Mechanism: For a batch of b pathology reports, the input can be modeled as $\mathbf{H}^t = \{\mathbf{H}_1^t, \mathbf{H}_2^t \dots \mathbf{H}_b^t\} \in \mathbb{R}^{b \times N_{t,i} \times d}$ where $N_{t,i}$ is the number of text segments in the ith report. Since reports vary in length, they are padded to ensure uniformity within the batch. The resulting padded batch is represented as: $\tilde{\mathbf{H}}^t = \{\tilde{\mathbf{H}}_1^t, \tilde{\mathbf{H}}_2^t \dots \tilde{\mathbf{H}}_b^t\} \in \mathbb{R}^{b \times m \times d_t}$ where m is the length of the longest report in the training set. A corresponding binary mask $\mathcal{M} \in \{0,1\}^{b \times m}$ is created to differentiate between the real tokens and padded positions, ensuring they do not contribute to the attention scores.

We aim to standardize report representations while preserving the most clinically and pathologically relevant information. To achieve this, we employ a Transformer-based self-attention mechanism [60] to construct the Diagnostic Prototypes. In particular, we project the padded report embeddings into query, key and value vectors using learnable linear transformations:

$$\mathbf{Q}^t = \mathbf{W}_Q \cdot \tilde{\mathbf{H}}^t, \quad \mathbf{K}^t = \mathbf{W}_K \cdot \tilde{\mathbf{H}}^t, \quad \mathbf{V}^t = \mathbf{W}_V \cdot \tilde{\mathbf{H}}^t \quad (1)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_t \times d_t}$ are learnable matrices. The scaled dot-product attention A^t is then computed as:

$$\mathbf{A}^t = \sigma\left(\frac{\mathbf{Q}^t \mathbf{K}^{t\top}}{\sqrt{d_t}}\right) \quad (2)$$

The padded positions are masked during the self-attention process to prevent them from influencing the results. The attention weights are further used to compute a weighted sum of the values $\mathbf{Z}^t = \mathbf{A}^t \cdot \mathbf{V}^t \in \mathbb{R}^{b \times m \times d_t}$ representing the post-attention embeddings of the pathology reports.

Prototype Selection: To extract the most diagnostically relevant segments (or prototypes) from each pathology report, we compute a single importance score for each segment, by averaging its attention weights across all query positions. This process produces a vector of importance scores for all sections in a report: $\mathbf{s}^t = [s_1^t, s_2^t, \cdots s_m^t] \in \mathbb{R}^m$, showing, on average, how much the entire report attends to each segment. Mathematically, if $\mathbf{A}_n^t \in \mathbb{R}^{m \times m}$ is the attention matrix for the n-th report, we define the importance score $s_{n,j}^t$ of section j as:

$$s_{n,j}^t = \frac{1}{m} \sum_{i=1}^{m} A_{n,i,j}^t \quad (3)$$

where $A_{n,i,j}^t$ is the attention weight from query position i to key position j. The sections are then ranked in descending order of their importance scores. Segments that are more relevant for diagnosis (*e.g.* those describing tumor size or pathology findings) typically receive higher scores. From the post-attention representation $\mathbf{Z}_i^t$, we select embeddings corresponding to the top $N_\mathcal{T}$ sections, yielding fixed-length Diagnostic Prototypes: $\mathbf{Z}_{i,\text{proto}}^t \in \mathbb{R}^{N_\mathcal{T} \times d_t}$, where $N_\mathcal{T}$ is the number of prototypes and d_t is the embedding dimension for the ith report. To facilitate multimodal fusion, we align the dimensions of the prototype representations derived from the three modalities. For this reason, we apply a linear transformation f_α^t to the text prototype embeddings, resulting in: $\mathbf{Z}_\alpha^t = f_\alpha^t(\mathbf{Z}_{\text{proto}}^t) \in \mathbb{R}^{d_e}$.

3.2. Histological prototypes from WSIs

Feature Extraction: For each WSI, we first identify and isolate the tissue regions [43] to ensure that diagnostically irrelevant background regions are excluded from further analysis. The retained tissue region in each histology image (h), denoted as X_i, is divided into non-overlapping patches [36] as follows: $X_i = \{x_1^h, x_2^h \dots x_{N_{h,i}}^h\}$ where $x_j^h \in \mathbb{R}^{H \times W \times 3}$ denotes the jth patch, H and W represent the height and width of the patch, respectively and $N_{h,i}$ represents the total number of patches in the ith slide. Each patch is processed by a pre-trained and frozen image encoder E_I to extract visual features: $\mathbf{h}_j^h = E^I(x_j^h) \in \mathbb{R}^{d_h}$, where d_h represents the dimensions of the extracted visual

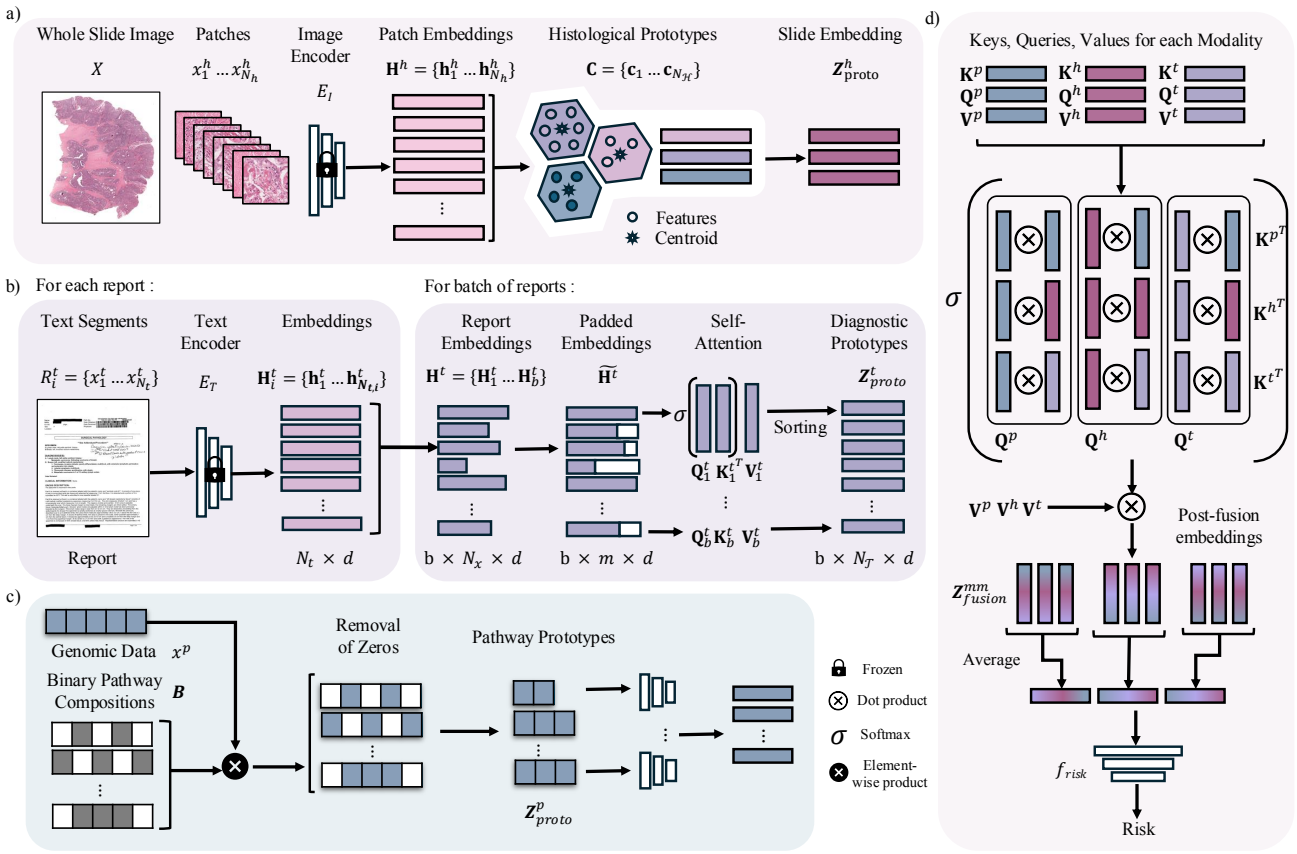

Figure 1. (a) Whole Slide Image X_i patches are processed by a frozen image encoder E_I to extract visual features $\mathbf{H}^h$, which are clustered into C histological prototypes using a Gaussian Mixture Model (GMM) to obtain a slide-level representation. (b) Feature embeddings $\mathbf{H}_i^t$ are extracted from the text segments of a single pathology report R_i, using a frozen text encoder E_T. For a batch of pathology reports $\mathbf{H}^t$ we construct Diagnostic prototypes by applying Transformer-based self-attention mechanisms and sorting the text segments within each report based on their importance scores. (c) Gene-expression vector $\mathbf{x}^p$ are multiplied with the binary pathway vectors ($\mathbf{B}$) to create pathway prototypes. These are processed through f_α MLPs to generate feature embeddings. (d) Prototypes from each modality are projected into key, value and query representations, followed by self- and cross-modal attention to integrate information. The resulting outputs are used for survival risk prediction.

feature vector. The set of extracted feature vectors for slide i are collectively represented as $\mathbf{H}_i^h = \{\mathbf{h}_1^h, \mathbf{h}_2^h \dots \mathbf{h}_{N_{h,i}}^h\}$.

Prototype Construction: To summarize the recurring morphological patterns in WSIs, we cluster patches into C histological prototypes using GMM, inspired by previous methods [50, 51]. Before applying the GMM, we first obtain an initial estimate of the histological prototypes by randomly initializing cluster centers in the feature space. Each prototype mean (μ) is sampled from a Gaussian distribution, ensuring that the cluster centers are spread out. The diagonal covariance matrices (Σ) are initialized as identity matrices while the mixture weights (π) are uniformly initialized. Mathematically we represent this as:

$$\mu_c \sim \mathcal{N}(0, 0.1^2 I), \quad \Sigma_c = I, \quad \pi_c = \frac{1}{N_{\mathcal{H}}} \tag{4}$$

$\forall c \in \{1, 2, \dots, N_{\mathcal{H}}\}$ where $N_{\mathcal{H}}$ denotes the number of prototypes c. Unlike traditional clustering methods, GMM allows soft assignments, meaning each patch is assigned to multiple prototypes with varying probabilities, enabling a more flexible representation of the morphological diversity found in the tissue structures [50]. The likelihood of a patch embedding under the GMM is given by:

$$p(\mathbf{h}_j^h) = \sum_{c=1}^{N_{\mathcal{H}}} \pi_c \cdot \mathcal{N}(\mathbf{h}_j^h; \mu_c, \Sigma_c) \tag{5}$$

where: π_c is the probability of selecting prototype c and $\mathcal{N}(\mathbf{h}_j^h; \mu_c, \Sigma_c)$ represents the Gaussian density function and models the likelihood of $\mathbf{h}_j^h$ given prototype c. For an entire

WSI, the joint probability over all patches is:

$$p(\mathbf{H}^h) = \prod_{j=1}^{N_h} \sum_{c=1}^{N_{\mathcal{H}}} \pi_c \cdot \mathcal{N}(\mathbf{h}_j^h; \mu_c, \Sigma_c) \tag{6}$$

This formulation ensures that the WSI is summarized using a compact set of histological prototypes. The GMM parameters (μ, Σ, π) are optimized using the Expectation-Maximization (EM) algorithm [8, 25]. Over successive EM iterations, the prototypes are progressively updated, ultimately converging to meaningful cluster centers. After convergence, the final slide representation is obtained by concatenating the estimated mixture parameters:

$$\mathbf{Z}^h_{\text{proto}} = [\pi_1, \mu_1, \Sigma_1, \ldots, \pi_{N_{\mathcal{H}}}, \mu_{N_{\mathcal{H}}}, \Sigma_{N_{\mathcal{H}}}] \in \mathbb{R}^{N_{\mathcal{H}} \times (1+2d_h)} \tag{7}$$

where: π_c quantifies the prevalence of each prototype in the slide, μ_c represents the average morphological characteristics of each prototype and $\sum_c$ captures the variation within the patches belonging to each prototype. This prototype-based embedding significantly reduces the dimensionality of WSI patch features whilst still preserving key morphological characteristics. To align the dimensionality of histology prototypes with those from other modalities, we apply a linear transformation f^h_α to the prototype embeddings, yielding : $\mathbf{Z}^h_\alpha = f^h_\alpha(\mathbf{Z}^h_{\text{proto}}) \in \mathbb{R}^{d_e}$.

3.3. Pathway Prototypes from Genomic Data

Prototype Construction: The transcriptomic profile for each sample can be represented by a gene-expression vector $\mathbf{x}^p \in \mathbb{R}^{N_G}$, where N_G is the total number of genes. We aim to group these genes (p) into $N_{\mathcal{P}}$ pre-defined biological pathways, each acting as a prototype. For each pathway i we define a binary mask $\mathbf{B}_i \in \{0,1\}^{N_G}$, where each element indicates whether a gene is included in a given pathway (1) or not (0). To construct the pathway prototypes, we perform an element-wise product of the gene expression vector $(\mathbf{x}^p)$ with the binary pathway vectors $(\mathbf{B})$ [51]. This operation generates a pathway-specific representation:

$$\mathbf{Z}^p_{\text{proto},i} = \mathbf{x}^p \odot \mathbf{B}_i, \quad \forall i \in [1, N_{\mathcal{P}}] \tag{8}$$

where $N_{\mathcal{P}}$ denotes the number of pathways. The resulting vector $\mathbf{Z}^p_{\text{proto},i}$ is then reduced to remove any zero entries, yielding a dense but variable-length representation (since different pathways involve different gene counts). To ensure a fixed size embedding for each pathway, we employ self-normalizing neural networks (SNNs) [20, 26] :

$$\mathbf{Z}^p_{i,\alpha} = f^p_{\alpha,i}(\mathbf{Z}^p_{\text{proto},i}) \in \mathbb{R}^{d_e}, \forall i \in [1, N_{\mathcal{P}}] \tag{9}$$

where $f^p_{\alpha,i}$ is a trainable network dedicated to pathway i. By treating each pathway as a "prototype" we are able to preserve biologically relevant groupings of genes while ensuring that the final representation for each pathway shares the same dimensionality with itself and aligns with the other modalities.

3.4. Multimodal Fusion

We aim to design a multimodal fusion model that extends the standard Transformer attention mechanism [10, 60] to capture both intra-modal and cross-modal interactions. Specifically, we seek to learn interactions among: diagnostic prototypes from pathology reports (t) represented as $\mathbf{Z}^t_\alpha \in \mathbb{R}^{N_{\mathcal{T}}}$, histological prototypes from histology images (h) represented as $\mathbf{Z}^h_\alpha \in \mathbb{R}^{N_{\mathcal{H}}}$ and pathway prototypes from transcriptomic data (p) represented as $\mathbf{Z}^p_\alpha \in \mathbb{R}^{N_{\mathcal{P}}}$. Each modality-specific embedding consists of $N_{\mathcal{M}}$ prototypes, where $\mathcal{M} \in \{\mathcal{P}, \mathcal{H}, \mathcal{T}\}$. To allow the model to dynamically enhance the prototype representations, we also append a learnable and randomly initialized embedding $\mathbf{e}_r \in \mathbb{R}^{d_r}$ to the prototypes [19, 32, 51]. The modified embeddings are then represented as $\mathbf{Z}^{h,p,t}_\alpha \in \mathbb{R}^{d=d_e+d_r}$.

To integrate the three modalities, we concatenate the modality-specific prototypes into a single multimodal embedding: $\mathbf{Z}^{mm} = [\mathbf{Z}^p_\alpha \parallel \mathbf{Z}^h_\alpha \parallel \mathbf{Z}^t_\alpha] \in \mathbb{R}^{(N_{\mathcal{P}}+N_{\mathcal{H}}+N_{\mathcal{T}})\times d}$, where d is the feature dimension. Following the standard Transformer architecture, we introduce three learnable projection matrices, $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d}$ to map the multimodal embedding $\mathbf{Z}^{mm}$ into queries, keys and values:

$$\mathbf{Q} = \mathbf{W}_Q \cdot \mathbf{Z}^{mm}, \quad \mathbf{K} = \mathbf{W}_K \cdot \mathbf{Z}^{mm}, \quad \mathbf{V} = \mathbf{W}_V \cdot \mathbf{Z}^{mm} \tag{10}$$

However, rather than treating $\mathbf{Z}^{mm}$ as a unified sequence, we decompose it into modality-specific components and compute attention for all possible pairwise interactions. Concretely, we split each $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ into three modality-specific blocks:

$$\begin{aligned} \mathbf{Q} &= \left[\mathbf{Q}^p \parallel \mathbf{Q}^h \parallel \mathbf{Q}^t\right], \\ \mathbf{K} &= \left[\mathbf{K}^p \parallel \mathbf{K}^h \parallel \mathbf{K}^t\right], \\ \mathbf{V} &= \left[\mathbf{V}^p \parallel \mathbf{V}^h \parallel \mathbf{V}^t\right], \end{aligned} \tag{11}$$

where $\mathbf{Q}^p, \mathbf{K}^p, \mathbf{Q}^p \in \mathbb{R}^{N_{\mathcal{P}}\times d}$ for pathways (and likewise for histology and text).

The approach results in a total of nine attention mechanisms categorized into (1) three self-attention (intra-modal) interactions: $\mathbf{A}^{p\to p}$ (pathways attending to pathways) , $\mathbf{A}^{h\to h}$(histology attending to histology) and $\mathbf{A}^{t\to t}$ (text attending to text) (2) six cross-attention (inter-model) interactions: $\mathbf{A}^{p\to h}, \mathbf{A}^{p\to t}$ (pathways attending to histology and text), $\mathbf{A}^{h\to p}, \mathbf{A}^{h\to t}$ (histology attending to pathways and text) and $\mathbf{A}^{t\to p}, \mathbf{A}^{t\to h}$ (text attending to pathways and histology). Each attention component is computed as: $\mathbf{A}^{m\to n} = \mathbf{Q}^m(\mathbf{K}^n)^\top$ where $\mathbf{A}^{m\to n}$ represents the attention logits from modality m to modality n.

After computing all intra- and cross-modal attention scores, we concatenate the sub-blocks within each modality and apply a row-wise softmax operation, producing a single probability distribution over key tokens for each query modality. This ensures that each modality attends to the

most relevant information from both itself and the other modalities. Once the attention scores are computed, they are multiplied by the corresponding value representations from all modalities. The final modality-specific embeddings after fusion are:

$$\begin{pmatrix} \mathbf{Z}^p_{\text{fusion}} \\ \mathbf{Z}^h_{\text{fusion}} \\ \mathbf{Z}^t_{\text{fusion}} \end{pmatrix} = \sigma \left[\frac{1}{\sqrt{d}} \begin{pmatrix} \mathbf{A}^{p\to p} & \mathbf{A}^{p\to h} & \mathbf{A}^{p\to t} \\ \mathbf{A}^{h\to p} & \mathbf{A}^{h\to h} & \mathbf{A}^{h\to t} \\ \mathbf{A}^{t\to p} & \mathbf{A}^{t\to h} & \mathbf{A}^{t\to t} \end{pmatrix} \right] \begin{pmatrix} \mathbf{V}^p \\ \mathbf{V}^h \\ \mathbf{V}^t \end{pmatrix}$$
$$= \sigma \left[\frac{1}{\sqrt{d}} \begin{pmatrix} \mathbf{Q}^p(\mathbf{K}^p)^\top & \mathbf{Q}^p(\mathbf{K}^h)^\top & \mathbf{Q}^p(\mathbf{K}^t)^\top \\ \mathbf{Q}^h(\mathbf{K}^p)^\top & \mathbf{Q}^h(\mathbf{K}^h)^\top & \mathbf{Q}^h(\mathbf{K}^t)^\top \\ \mathbf{Q}^t(\mathbf{K}^p)^\top & \mathbf{Q}^t(\mathbf{K}^h)^\top & \mathbf{Q}^t(\mathbf{K}^t)^\top \end{pmatrix} \right] \begin{pmatrix} \mathbf{V}^p \\ \mathbf{V}^h \\ \mathbf{V}^t \end{pmatrix} \tag{12}$$

where $\sigma[\cdot]$ is the row-wise softmax operation. Finally, the full multimodal representation is obtained by concatenating the post-fusion modality-specific embeddings:

$$\mathbf{Z}^{mm}_{\text{fusion}} = [\mathbf{Z}^p_{\text{fusion}} \| \mathbf{Z}^h_{\text{fusion}} \| \mathbf{Z}^t_{\text{fusion}}] \tag{13}$$

3.5. Survival Prediction

After the multimodal fusion step, the resulting embeddings are processed by a series of multi-layer perceptrons (MLPs) for further transformation. Within each modality, these embeddings undergo layer normalization and are subsequently averaged to create compact, modality-specific representations [51].

$$risk = f_{risk} \left[\frac{1}{N_{\mathcal{X}}} \sum_{i=1}^{N_{\mathcal{X}}} f_\beta(\mathbf{Z}^x_{i,\text{fusion}}) \right]_{\mathcal{X}\in\{\mathcal{P},\mathcal{H},\mathcal{T}\}} \tag{14}$$

The final patient-level representation is obtained by concatenating these averaged modality-specific embeddings, which are then passed through a linear layer f_{risk} to generate a patient-level prediction.

4. Datasets and Implementation

4.1. Datasets

We evaluate our proposed method using publicly available datasets from The Cancer Genome Atlas (TCGA) across six types of cancer [65]: Bladder urothelial carcinoma (BLCA) (n = 328), Lung adenocarcinoma (LUAD) (n = 391), Kidney renal clear cell carcinoma (KIRC) (n = 328), Stomach adenocarcinoma (STAD) (n = 253), Colon and Rectum adenocarcinoma (CRC) (n = 269) and Head and Neck Squamous Cell Carcinoma (HNSC) (n= 385).

Hematoxylin and Eosin (H&E) stained WSIs for all patients were obtained from the National Cancer Institute (NCI) Genomic Data Commons (GDC) [65]. The corresponding pathology reports were obtained from Kefeli *et al*.[23], who curated 9,523 machine-readable reports from TCGA by converting PDF documents into text format. Bulk RNA sequencing data for TCGA cohorts was accessed from the UCSC Xena database [14], measured using the Illumina HiSeq 2000 system. To stabilize variance and mitigate the impact of extreme values, we applied and $log_2(x+1)$ transformed with RSEM normalization [28]. To structure the data into pathways, we utilized 50 Hallmark gene sets from the Molecular Signatures Database (MSigDB) [33, 55], covering approximately 4000 unique genes in total. Further implementation details can be found in Supplementary Material, **Implementation Details**.

4.2. Baselines

4.2.1. Unimodal Baselines

The following unimodal baselines were separately implemented for each data modality:

Text: We employ a unimodal text-only variant of the proposed method (with prototypes), PS3_t. Additionally, we use an Attention-based MIL [18] on the pathology reports. **Gene:** We use a unimodal transcriptomics-only variant of the proposed method (with prototypes), PS3_p. Additionally, we use a 2-layer MLP as a non-prototype baseline [20, 51]. **Histology:** We utilize TransMIL [48], R^2-TMIL [57] and CLAM [36, 69] as our histology baselines. We also implement a unimodal histology-only variant of the proposed method (with prototypes), PS3_WSI.

4.2.2. Multimodal Baselines

We implement MOTCat [67], MCAT [5], SurvPath [20], SurvivMIL [40], CMTA [74], FSM [73] and both variants of MMP [51] - MMP_Trans and MMP_OT — as our multimodal baselines. Additional details can be found in Supplementary Material, **Multimodal Baselines**.

5. Results

5.1. Survival Prediction Results

Table.1 presents the C-Index results for PS3 and all baseline models in predicting disease-specific survival. PS3 demonstrates the highest overall (average) performance, outperforming the next-best multimodal and unimodal methods with percentage improvements of 6.72% and 7.21%, respectively. In addition, it achieves among the highest performance in 5 of the 6 cancer types evaluated. We summarize the key findings below. Additional results can be found in Supplementary Material, **Kaplan-Meier Analysis** and **Attention Visualization**.

5.1.1. Comparison with Clinical Baseline

Clinical variables such as age, sex and histologic grade have been identified as important prognostic factors for survival prediction [22, 56, 58, 61]. In Table.1 our analysis demonstrates that all subsequent methods achieve superior perfor-

	Model	BLCA	LUAD	KIRC	STAD	CRC	HNSC	Avg (↑)
	Clinical	0.557 ± 0.062	0.494 ± 0.093	0.723 ± 0.044	0.583 ± 0.051	0.496 ± 0.099	0.516 ± 0.090	0.561
Gene	Gene exp [20]	0.656 ± 0.047	0.508 ± 0.064	0.764 ± 0.042	0.578 ± 0.080	0.678 ± 0.069	0.595 ± 0.076	0.630
	$PS3_p$	0.643 ± 0.062	0.578 ± 0.068	0.765 ± 0.040	0.614 ± 0.059	0.710 ± 0.063	0.601 ± 0.060	0.652
Text	Text ABMIL [18]	0.522 ± 0.069	0.560 ± 0.053	0.598 ± 0.079	0.594 ± 0.067	0.772 ± 0.137	0.507 ± 0.067	0.592
	$PS3_t$	0.629 ± 0.065	0.536 ± 0.087	0.623 ± 0.075	0.548 ± 0.078	0.805 ± 0.130	0.550 ± 0.080	0.615
Histology	CLAM [36]	0.562 ± 0.100	0.606 ± 0.104	0.684 ± 0.055	0.510 ± 0.078	0.680 ± 0.127	0.560 ± 0.075	0.600
	TransMIL [48]	0.585 ± 0.049	0.594 ± 0.096	0.736 ± 0.085	0.558 ± 0.035	0.671 ± 0.120	0.594 ± 0.070	0.623
	RRTMIL [57]	0.550 ± 0.065	0.557 ± 0.091	0.704 ± 0.128	0.603 ± 0.088	0.575 ± 0.035	0.556 ± 0.056	0.591
	$PS3_{WSI}$	0.539 ± 0.033	0.597 ± 0.071	0.759 ± 0.102	0.542 ± 0.085	0.586 ± 0.197	0.502 ± 0.053	0.588
Multimodal	SurvivMIL [40]	0.484 ± 0.050	0.568 ± 0.048	0.654 ± 0.101	0.492 ± 0.068	0.663 ± 0.057	0.552 ± 0.086	0.569
	SurvPath [20]	0.613 ± 0.036	0.567 ± 0.055	0.761 ± 0.054	0.608 ± 0.048	0.640 ± 0.054	0.536 ± 0.055	0.621
	MOTCat [67]	0.636 ± 0.057	0.533 ± 0.039	0.766 ± 0.049	0.553 ± 0.082	0.677 ± 0.067	0.586 ± 0.044	0.625
	MCAT [5]	0.636 ± 0.068	0.512 ± 0.040	0.762 ± 0.030	0.572 ± 0.074	0.661 ± 0.101	0.578 ± 0.064	0.620
	CMTA [74]	0.637 ± 0.067	0.565 ± 0.045	0.741 ± 0.044	0.578 ± 0.065	0.659 ± 0.058	0.579 ± 0.030	0.627
	FSM [73]	0.642 ± 0.050	0.565 ± 0.082	$\mathbf{0.776} \pm 0.048$	0.609 ± 0.068	0.663 ± 0.059	0.577 ± 0.064	0.639
	MMP_{Trans} [51]	0.641 ± 0.053	0.606 ± 0.068	$\mathbf{0.776} \pm 0.059$	0.639 ± 0.063	0.689 ± 0.078	0.566 ± 0.075	0.653
	MMP_{OT} [51]	0.645 ± 0.030	0.617 ± 0.058	0.774 ± 0.026	$\mathbf{0.660} \pm 0.073$	0.689 ± 0.074	0.545 ± 0.041	0.655
	PS3	$\mathbf{0.684} \pm 0.026$	$\mathbf{0.640} \pm 0.093$	$\mathbf{0.776} \pm 0.061$	0.638 ± 0.045	$\mathbf{0.826} \pm 0.101$	$\mathbf{0.627} \pm 0.066$	**0.699**

Table 1. Survival prediction results comparing the proposed method, PS3, with multimodal and unimodal baselines for disease-specific survival prediction using the C-Index. Pathology Language and Image Pre-Training (PLIP) [16] is used as a feature encoder across all methods. Performance is evaluated over five runs, with standard deviation reported. The best-performing results are highlighted in bold.

mance (by a minimum of 1.4%) compared to the clinical baseline, which relies solely on these variables. This highlights the benefits of integrating additional data modalities such as histology images, transcriptomic data and pathology reports. Additional details can be sound in Supplementary Material, **Clinical Baselines**.

Ablation	Model	Avg.
	PS3	**0.699**
Text Proto $N_{\mathcal{T}}$	Avg ⇒ p90	0.691 (-1.14%)
Encoder E_I, E_T	PLIP ⇒ QUILT-Net	0.676 (-3.29%)
Modalities	$p, h, t \Rightarrow h, t$	0.644 (-7.87%)
	$p, h, t \Rightarrow p, t$	0.687 (-1.72%)
	$p, h, t \Rightarrow p, h$	0.653 (-6.58%)
Fusion Method	Full ⇒ Late	0.669 (-4.29%)
	Full ⇒ Hierarchical	0.660 (-5.58%)
Embeddings $\mathbf{e}_r$	Random ⇒ None	0.688 (-1.57%)
MLP f_β	Multiple ⇒ Single	0.690 (-1.29%)

Table 2. Ablation study analyzing the impact of modifying individual model components on the C-Index, with results averaged across six cohorts.

5.1.2. Comparison of Histology, Gene and Text-Based Methods

Table.1 also demonstrates that gene-based methods (with C-Index results $0.63 - 0.65$) outperform other unimodal baselines, highlighting the strong prognostic value of transcriptomics data for survival prediction. In contrast, histology- (with C-Index results $0.58 - 0.62$) and text-based methods (with C-Index results $0.59 - 0.61$) achieve similar performance, likely due to their shared reliance on histopathological features. While WSIs provide direct morphological insights and pathology reports summarize these observations, both modalities may be more affected by feature extraction challenges and variability in reporting, making them less discriminative than genomic features.

5.1.3. Impact of Prototypes in Unimodal and Multimodal Approaches

In unimodal settings, prototype-based methods outperform non-prototype approaches for gene and text data. However, for histology, non-prototype methods achieve higher performance, as seen in Table.1, where TransMIL, CLAM and R^2-TMIL outperform $PS3_{WSI}$. This decline can be attributed to the compression effect of histological prototyping. Reducing thousands of image patches to a compact set of prototypes reduces data dimensionality and complexity but can lead to the loss of fine-grained histological details.

However, prototyping can enhance multimodal learning by facilitating better integration across modalities. While

MIL-based methods perform well in unimodal settings, they tend to struggle with multimodal integration. In contrast, prototyping-based approaches significantly outperform MIL-based methods when combining histology with transcriptomics and/or text. As shown in Table.1, PS3, $\text{MMP}_{\text{Trans}}$ and MMP_{OT} outperform SurvPath, MCAT, CMTA and MOTCat, demonstrating the benefits of prototyping for multimodal fusion. A similar trend is observed in the histology-text configuration where the histology + text method (h, t) ablation study in Table.2 outperforms MIL-based method, SurvivMIL by 13.18%. Data scale disparities across modalities can lead to modality dominance and imbalance, hindering effective multimodal integration. Prototyping mitigates this by normalizing data scale while preserving essential information, ensuring more balanced multimodal integration where each modality contributes meaningfully to survival prediction.

5.1.4. Two-Modal vs. Three-Modal Approaches

As shown in Table.1, integrating all three modalities outperforms both dual-modality and unimodal approaches, achieving 6.72% and 7.21% improvement compared to the next-best multimodal (with C-Index 0.655) and unimodal (with C-Index 0.652) methods, respectively. This underscores the value of incorporating pathology reports into the workflow. Further supporting this, Table.2 presents an ablation study (Impact of Modality Combinations), which evaluates our model's performance across different two-modality combinations. The results show that the proposed three-modal approach, PS3 $(h, p, t - 0.699)$ achieves 8.54% higher performance than Histology + Text $(h, t - 0.644)$, 1.75% higher performance than Pathways + Text $(p, t - 0.687)$ and 7.04% higher performance than Pathways + Histology $(p, h - 0.653)$. Notably, Pathways + Text outperforms both other dual-modality combinations, highlighting the prognostic value of pathology reports. These results highlight the advantage of multimodal integration in improving patient prognostication.

5.2. Ablation Study

We conducted extensive ablation studies in Table.2 to evaluate the impact of various design choices on multimodal learning performance. Below, we summarize our key findings: (1) **Number of Diagnostic Text Prototypes :** Instead of setting the number of text prototypes $N_{\mathcal{T}}$ to the average length of text segments in the training dataset, we experimented with setting it to the 90th percentile (p90). However, this introduced additional noise, which reduced the overall performance. (2) **Feature Encoder Performance:** We compared different vision-language feature encoders and discovered that PLIP-based [16] features outperformed those derived from QUILT-Net [17]. (3) **Impact of Modality Combinations:** As explained previously in Section.5.1.4 the proposed model which incorporates all three modalities (h, p, t) outperforms all two-modality combinations, highlighting the importance of integrating multiple complementary data sources. (4) **Fusion Methods Comparison:** We evaluated two different fusion strategies: *Late Fusion*: This approach only applies attention within each modality, without capturing any cross-modal interactions. Modalities are only merged at the final stage before prediction. *Hierarchical Fusion*: This approach employs a two-step process. First, it models cross- and self-modal interactions between histology and text. Then, the fused (histology + text) representation undergoes additional self and cross-attention with transcriptomics. We note that the proposed method outperforms both aforementioned fusion strategies. (5) **Effect of Learnable Embeddings:** We experimented without adding the randomly initialized embeddings, $\mathbf{e}_r$ to the feature representations of each modality. Our results indicate that these embeddings improve model performance by allowing the model to learn richer feature representations. (6) **MLPs:** Instead of separate MLPs for the prototypes we experiment with implementing a single shared MLP for all prototypes across different modalities.

6. Conclusions

Pathology reports contain critical diagnostic and prognostic insights, yet remain under-utilized in computational survival prediction models. While transcriptomic and histological data capture molecular and spatial characteristics, pathology reports offer expert-driven interpretations that complement these modalities. However, existing approaches often overlook this valuable resource in multimodal survival analysis. To address this, we proposed a prototype-based multimodal fusion framework that integrates cancer aggressiveness-related signals from WSIs, pathology reports and transcriptomic data for improved survival prediction. By transforming each modality into structured representations – histological prototypes for WSIs, diagnostic prototypes for pathology reports and pathway prototypes for transcriptomic data – we mitigated modality imbalance and improved predictive performance. The PS3 multimodal transformer effectively modeled cross-modal interactions across all three modalities, leading to superior predictive performance. We evaluated our proposed model across six cancer types, outperforming both unimodal and multimodal baselines.

Future work can further enhance this approach by incorporating additional multi-omics modalities, radiology images and patient metadata, to provide a more comprehensive view of patient characteristics. By continuing to refine multimodal fusion strategies, we can advance precision oncology and improve patient outcomes.

Acknowledgments

The first author would like to acknowledge support from AstraZeneca, UK and the Department of Computer Science at the University of Warwick, UK.

References

[1] Sajjad Abedian, Evan T Sholle, Prakash M Adekkanattu, Marika M Cusick, Stephanie E Weiner, Jonathan E Shoag, Jim C Hu, and Thomas R Campion Jr. Automated extraction of tumor staging and diagnosis information from surgical pathology reports. *JCO clinical cancer informatics*, 5: 1054–1061, 2021. 1

[2] Martim Afonso, Praphulla MS Bhawsar, Monjoy Saha, Jonas S Almeida, and Arlindo L Oliveira. Multiple instance learning for wsi: A comparative analysis of attention-based approaches. *Journal of Pathology Informatics*, 15:100403, 2024. 2

[3] Richard J Chen, Ming Y Lu, Jingwen Wang, Drew FK Williamson, Scott J Rodig, Neal I Lindeman, and Faisal Mahmood. Pathomic fusion: an integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis. *IEEE Transactions on Medical Imaging*, 41(4):757–770, 2020. 2

[4] Richard J Chen, Ming Y Lu, Muhammad Shaban, Chengkuan Chen, Tiffany Y Chen, Drew FK Williamson, and Faisal Mahmood. Whole slide images are 2d point clouds: Context-aware survival prediction using patch-based graph convolutional networks. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part VIII 24*, pages 339–349. Springer, 2021. 2

[5] Richard J Chen, Ming Y Lu, Wei-Hung Weng, Tiffany Y Chen, Drew FK Williamson, Trevor Manz, Maha Shady, and Faisal Mahmood. Multimodal co-attention transformer for survival prediction in gigapixel whole slide images. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4025, 2021. 1, 3, 6, 7

[6] Richard J Chen, Ming Y Lu, Drew FK Williamson, Tiffany Y Chen, Jana Lipkova, Zahra Noor, Muhammad Shaban, Maha Shady, Mane Williams, Bumjin Joo, et al. Pan-cancer integrative histology-genomic analysis via multimodal deep learning. *Cancer cell*, 40(8):865–878, 2022. 1, 2

[7] P Deepa and C Gunavathi. A systematic review on machine learning and deep learning techniques in cancer survival prediction. *Progress in Biophysics and Molecular Biology*, 174: 62–71, 2022. 1

[8] Arthur P Dempster, Nan M Laird, and Donald B Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the royal statistical society: series B (methodological)*, 39(1):1–22, 1977. 5

[9] Kexin Ding, Mu Zhou, Dimitris N Metaxas, and Shaoting Zhang. Pathology-and-genomics multimodal transformer for survival outcome prediction. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 622–631. Springer, 2023. 3

[10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 2, 5

[11] Michael Gadermayr and Maximilian Tschuchnig. Multiple instance learning for digital pathology: A review of the state-of-the-art, limitations & future potential. *Computerized Medical Imaging and Graphics*, page 102337, 2024. 2

[12] Konrad Gadzicki, Razieh Khamsehashari, and Christoph Zetzsche. Early vs late fusion in multimodal convolutional neural networks. In *2020 IEEE 23rd international conference on information fusion (FUSION)*, pages 1–6. IEEE, 2020. 1, 2

[13] Thomas M Gill. The central role of prognosis in clinical decision making. *Jama*, 307(2):199–200, 2012. 1

[14] Mary J Goldman, Brian Craft, Mim Hastie, Kristupas Repečka, Fran McDade, Akhil Kamath, Ayan Banerjee, Yunhai Luo, Dave Rogers, Angela N Brooks, et al. Visualizing and interpreting cancer genomics data via the xena platform. *Nature biotechnology*, 38(6):675–678, 2020. 6

[15] Shih-Cheng Huang, Anuj Pareek, Saeed Seyyedi, Imon Banerjee, and Matthew P Lungren. Fusion of medical imaging and electronic health records using deep learning: a systematic review and implementation guidelines. *NPJ digital medicine*, 3(1):136, 2020. 2

[16] Zhi Huang, Federico Bianchi, Mert Yuksekgonul, Thomas J Montine, and James Zou. A visual–language foundation model for pathology image analysis using medical twitter. *Nature medicine*, 29(9):2307–2316, 2023. 7, 8

[17] Wisdom Ikezogwo, Saygin Seyfioglu, Fatemeh Ghezloo, Dylan Geva, Fatwir Sheikh Mohammed, Pavan Kumar Anand, Ranjay Krishna, and Linda Shapiro. Quilt-1m: One million image-text pairs for histopathology. *Advances in neural information processing systems*, 36, 2024. 8

[18] Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *International conference on machine learning*, pages 2127–2136. PMLR, 2018. 2, 6, 7

[19] Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, Carl Doersch, Catalin Ionescu, David Ding, Skanda Koppula, Daniel Zoran, Andrew Brock, Evan Shelhamer, et al. Perceiver io: A general architecture for structured inputs & outputs. *arXiv preprint arXiv:2107.14795*, 2021. 5

[20] Guillaume Jaume, Anurag Vaidya, Richard J Chen, Drew FK Williamson, Paul Pu Liang, and Faisal Mahmood. Modeling dense multimodal interactions between biological pathways and histology for survival prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11579–11590, 2024. 1, 2, 3, 5, 6, 7

[21] Shuai Jiang, Arief A Suriawinata, and Saeed Hassanpour. Mhattnsurv: Multi-head attention for survival prediction using whole-slide pathology images. *Computers in biology and medicine*, 158:106883, 2023. 2

[22] Arjen Joosse, Sandra Collette, Stefan Suciu, Tamar Nijsten, Poulam M Patel, Ulrich Keilholz, Alexander MM Eggermont, Jan Willem W Coebergh, and Esther de Vries.

Sex is an independent prognostic indicator for survival and relapse/progression-free survival in metastasized stage iii to iv melanoma: a pooled analysis of five european organisation for research and treatment of cancer randomized controlled trials. *Journal of Clinical Oncology*, 31(18):2337–2346, 2013. 6

[23] Jenna Kefeli and Nicholas Tatonetti. Tcga-reports: A machine-readable pathology report resource for benchmarking text-based ai models. *Patterns*, 5(3), 2024. 1, 6

[24] Jenna Kefeli, Jacob Berkowitz, Jose M Acitores Cortina, Kevin K Tsang, and Nicholas P Tatonetti. Generalizable and automated classification of tnm stage from pathology reports with external validation. *Nature Communications*, 15 (1):8916, 2024. 1

[25] Minyoung Kim. Differentiable expectation-maximization for set representation learning. In *International Conference on Learning Representations*, 2022. 2, 5

[26] Günter Klambauer, Thomas Unterthiner, Andreas Mayr, and Sepp Hochreiter. Self-normalizing neural networks. *Advances in neural information processing systems*, 30, 2017. 5

[27] Jeongeun Lee, Hyun-Je Song, Eunsil Yoon, Seong-Bae Park, Sung-Hye Park, Jeong-Wook Seo, Peom Park, and Jinwook Choi. Automated extraction of biomarker information from pathology reports. *BMC medical informatics and decision making*, 18:1–11, 2018. 1

[28] Bo Li and Colin N Dewey. Rsem: accurate transcript quantification from rna-seq data with or without a reference genome. *BMC bioinformatics*, 12:1–16, 2011. 6

[29] Chunyuan Li, Xinliang Zhu, Jiawen Yao, and Junzhou Huang. Hierarchical transformer for survival prediction using multimodality whole slide images and genomics. In *2022 26th international conference on pattern recognition (ICPR)*, pages 4256–4262. IEEE, 2022. 3

[30] Ruoyu Li, Jiawen Yao, Xinliang Zhu, Yeqing Li, and Junzhou Huang. Graph cnn for survival analysis on whole slide pathological images. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 174–182. Springer, 2018. 2

[31] Ruiqing Li, Xingqi Wu, Ao Li, and Minghui Wang. Hfbsurv: hierarchical multimodal fusion with factorized bilinear models for cancer survival prediction. *Bioinformatics*, 38(9): 2587–2594, 2022. 1, 2

[32] Paul Pu Liang, Yiwei Lyu, Xiang Fan, Jeffrey Tsaw, Yudong Liu, Shentong Mo, Dani Yogatama, Louis-Philippe Morency, and Ruslan Salakhutdinov. High-modality multimodal transformer: Quantifying modality & interaction heterogeneity for high-modality representation learning. *arXiv preprint arXiv:2203.01311*, 2022. 5

[33] Arthur Liberzon, Chet Birger, Helga Thorvaldsdóttir, Mahmoud Ghandi, Jill P Mesirov, and Pablo Tamayo. The molecular signatures database hallmark gene set collection. *Cell systems*, 1(6):417–425, 2015. 2, 6

[34] Mingxin Liu, Yunzan Liu, Hui Cui, Chunquan Li, and Jiquan Ma. Mgct: Mutual-guided cross-modality transformer for survival outcome prediction using integrative histopathology-genomic features. In *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 1306–1312. IEEE, 2023. 3

[35] Pei Liu, Luping Ji, Feng Ye, and Bo Fu. Graphlsurv: A scalable survival prediction network with adaptive and sparse structure learning for histopathological whole-slide images. *Computer methods and programs in biomedicine*, 231:107433, 2023. 2

[36] Ming Y Lu, Drew FK Williamson, Tiffany Y Chen, Richard J Chen, Matteo Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature biomedical engineering*, 5(6):555–570, 2021. 3, 6, 7

[37] Zhilong Lv, Yuexiao Lin, Rui Yan, Ying Wang, and Fa Zhang. Transsurv: transformer-based survival analysis model integrating histopathological images and genomic data for colorectal cancer. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 20(6):3411–3420, 2022. 3

[38] William J Mackillop. The importance of prognosis in cancer medicine. *TNM Online*, 2003. 1

[39] Pooya Mobadersany, Safoora Yousefi, Mohamed Amgad, David A Gutman, Jill S Barnholtz-Sloan, José E Velázquez Vega, Daniel J Brat, and Lee AD Cooper. Predicting cancer outcomes from histology and genomics using convolutional networks. *Proceedings of the National Academy of Sciences*, 115(13):E2970–E2979, 2018. 2

[40] Reed Naidoo, Olga Fourkioti, Matt De Vries, and Chris Bakal. Survivmil: A multimodal, multiple instance learning pipeline for survival outcome of neuroblastoma patients. In *MICCAI Workshop on Computational Pathology with Multimodal Data (COMPAYL)*, 2024. 2, 6, 7

[41] Giulio Napolitano, Adele Marshall, Peter Hamilton, and Anna T Gavin. Machine learning classification of surgical pathology reports and chunk recognition for information extraction noise reduction. *Artificial intelligence in medicine*, 70:77–83, 2016. 1, 2

[42] Gil Patrus Pena and Joséde Souza Andrade-Filho. How does a pathologist make a diagnosis? *Archives of pathology & laboratory medicine*, 133(1):124–132, 2009. 1

[43] Johnathan Pocock, Simon Graham, Quoc Dang Vu, Mostafa Jahanifar, Srijay Deshpande, Giorgos Hadjigeorghiou, Adam Shephard, Raja Muhammad Saad Bashir, Mohsin Bilal, Wenqi Lu, et al. Tiatoolbox as an end-to-end library for advanced tissue image analytics. *Communications medicine*, 2 (1):120, 2022. 3

[44] Dario Pugliese, Giuseppe Palermo, Angelo Totaro, Pier Francesco Bassi, and Francesco Pinto. Clinical, pathological and molecular prognostic factors in prostate cancer decision-making process. *Urologia Journal*, 83(1): 14–20, 2016. 1

[45] Lin Qiu, Aminollah Khormali, and Kai Liu. Deep biological pathway informed pathology-genomic multimodal survival prediction. *arXiv preprint arXiv:2301.02383*, 2023. 2

[46] Manahil Raza, Ruqayya Awan, Raja Muhammad Saad Bashir, Talha Qaiser, and Nasir M Rajpoot. Dual attention model with reinforcement learning for classification of histology whole-slide images. *Computerized Medical Imaging and Graphics*, 118:102466, 2024. 2

[47] Thiago Santos, Amara Tariq, Judy Wawira Gichoya, Hari Trivedi, and Imon Banerjee. Automatic classification of cancer pathology reports: a systematic review. *Journal of Pathology Informatics*, 13:100003, 2022. 2

[48] Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, et al. Transmil: Transformer based correlated multiple instance learning for whole slide image classification. *Advances in neural information processing systems*, 34:2136–2147, 2021. 6, 7

[49] Zhuchen Shao, Yang Chen, Hao Bian, Jian Zhang, Guojun Liu, and Yongbing Zhang. Hvtsurv: Hierarchical vision transformer for patient-level survival prediction from whole slide image. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2209–2217, 2023. 2

[50] Andrew H Song, Richard J Chen, Tong Ding, Drew FK Williamson, Guillaume Jaume, and Faisal Mahmood. Morphological prototyping for unsupervised slide representation learning in computational pathology. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11566–11578, 2024. 2, 3, 4

[51] Andrew H Song, Richard J Chen, Guillaume Jaume, Anurag Jayant Vaidya, Alexander Baras, and Faisal Mahmood. Multimodal prototyping for cancer survival prediction. In *Forty-first International Conference on Machine Learning*, 2024. 1, 2, 3, 4, 5, 6, 7

[52] Sören Richard Stahlschmidt, Benjamin Ulfenborg, and Jane Synnergren. Multimodal deep learning for biomedical data fusion: a review. *Briefings in Bioinformatics*, 23(2):bbab569, 2022. 1

[53] Eric Steimetz, Elmira Mostafidi, Carolina Castagna, Raavi Gupta, and Rosemary Frasso. Forgotten clientele: A systematic review of patient-centered pathology reports. *Plos one*, 19(5):e0301116, 2024. 1

[54] Sandra Steyaert, Marija Pizurica, Divya Nagaraj, Priya Khandelwal, Tina Hernandez-Boussard, Andrew J Gentles, and Olivier Gevaert. Multimodal data fusion for cancer biomarker discovery with deep learning. *Nature machine intelligence*, 5(4):351–362, 2023. 1

[55] Aravind Subramanian, Pablo Tamayo, Vamsi K Mootha, Sayan Mukherjee, Benjamin L Ebert, Michael A Gillette, Amanda Paulovich, Scott L Pomeroy, Todd R Golub, Eric S Lander, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43):15545–15550, 2005. 6

[56] Zhifu Sun, Marie-Christine Aubry, Claude Deschamps, Randolph S Marks, Scott H Okuno, Brent A Williams, Hiroshi Sugimura, V Shane Pankratz, and Ping Yang. Histologic grade is an independent prognostic factor for survival in non–small cell lung cancer: An analysis of 5018 hospital-and 712 population-based cases. *The Journal of thoracic and cardiovascular surgery*, 131(5):1014–1020, 2006. 6

[57] Wenhao Tang, Fengtao Zhou, Sheng Huang, Xiang Zhu, Yi Zhang, and Bo Liu. Feature re-embedding: Towards foundation model-level performance in computational pathology. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11343–11352, 2024. 6, 7

[58] Faruk Tas, Rumeysa Ciftci, Leyla Kilic, and Senem Karabulut. Age is a prognostic factor affecting survival in lung cancer patients. *Oncology letters*, 6(5):1507–1513, 2013. 6

[59] Daniel Truhn, Chiara ML Loeffler, Gustav Müller-Franzes, Sven Nebelung, Katherine J Hewitt, Sebastian Brandner, Keno K Bressem, Sebastian Foersch, and Jakob Nikolas Kather. Extracting structured information from unstructured histopathology reports using generative pre-trained transformer 4 (gpt-4). *The Journal of Pathology*, 262(3):310–319, 2024. 2

[60] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. 2, 3, 5

[61] Myrella Vlenterie, Vincent KY Ho, Suzanne EJ Kaal, Richelle Vlenterie, Rick Haas, and Winette TA Van Der Graaf. Age as an independent prognostic factor for survival of localised synovial sarcoma patients. *British Journal of Cancer*, 113(11):1602–1606, 2015. 6

[62] Quoc Dang Vu, Kashif Rajpoot, Shan E Ahmed Raza, and Nasir Rajpoot. Handcrafted histological transformer (h2t): Unsupervised representation of whole slide images. *Medical image analysis*, 85:102743, 2023. 2

[63] Jun Wang, Yu Mao, Nan Guan, and Chun Jason Xue. Advances in multiple instance learning for whole slide image analysis: Techniques, challenges, and future directions. *arXiv preprint arXiv:2408.09476*, 2024. 2

[64] Zhiqin Wang, Ruiqing Li, Minghui Wang, and Ao Li. Gpdbn: deep bilinear network integrating both genomic data and pathological images for breast cancer prognosis prediction. *Bioinformatics*, 37(18):2963–2970, 2021. 1, 2

[65] John N Weinstein, Eric A Collisson, Gordon B Mills, Kenna R Shaw, Brad A Ozenberger, Kyle Ellrott, Ilya Shmulevich, Chris Sander, and Joshua M Stuart. The cancer genome atlas pan-cancer analysis project. *Nature genetics*, 45(10):1113–1120, 2013. 2, 6

[66] Peng Xu, Xiatian Zhu, and David A Clifton. Multimodal learning with transformers: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10):12113–12132, 2023. 3

[67] Yingxue Xu and Hao Chen. Multimodal optimal transport-based co-attention transformer with global structure consistency for survival prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21241–21251, 2023. 1, 3, 6, 7

[68] Rui Yan, Zhilong Lv, Zhidong Yang, Senlin Lin, Chunhou Zheng, and Fa Zhang. Sparse and hierarchical transformer for survival analysis on whole slide images. *IEEE Journal of Biomedical and Health Informatics*, 28(1):7–18, 2023. 2

[69] Zhaochang Yang, Ting Wei, Ying Liang, Xin Yuan, Ruitian Gao, Yujia Xia, Jie Zhou, Yue Zhang, and Zhangsheng Yu. A foundation model for generalizable cancer diagnosis and survival prediction from histopathological images. *bioRxiv*, pages 2024–05, 2024. 6

[70] Jiawen Yao, Xinliang Zhu, Jitendra Jonnagaddala, Nicholas Hawkins, and Junzhou Huang. Whole slide images based cancer survival prediction using attention guided deep multiple instance learning networks. *Medical image analysis*, 65: 101789, 2020. 2

[71] Wataru Yasui, Naohide Oue, Phyu Phyu Aung, Shunji Matsumura, Mariko Shutoh, and Hirofumi Nakayama. Molecular-pathological prognostic factors of gastric cancer: a review. *Gastric cancer*, 8:86–94, 2005. 1

[72] Yilan Zhang, Yingxue Xu, Jianqi Chen, Fengying Xie, and Hao Chen. Prototypical information bottlenecking and disentangling for multimodal cancer survival prediction. *arXiv preprint arXiv:2401.01646*, 2024. 1, 2, 3

[73] Yi Zheng, Regan D Conrad, Emily J Green, Eric J Burks, Margrit Betke, Jennifer E Beane, and Vijaya B Kolachalama. Graph attention-based fusion of pathology images and gene expression for prediction of cancer survival. *IEEE transactions on medical imaging*, 2024. 3, 6, 7

[74] Fengtao Zhou and Hao Chen. Cross-modal translation and alignment for survival analysis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21485–21494, 2023. 3, 6, 7